Overview

A5 series provide advanced control of AC induction motors and permanent magnet drive motors performing on-vehicle traction drive or hydraulic pump duties. They offer vehicle developers a highly cost- effective combination of power, performance and functionality.

The A5 series deployed with multiple patented techniques, that extend life of build-in relays and MOSFET units, a higher reliability is achieved.

Typical Application

Small and medium material handling, light on-road, pallet trucks, stacker, golf, personnel carriers.

Features

Full Featured

Offers 0~300Hz wide working frequency.

Available for 400A output at 36-72V system voltages. These are true 2 minute Arms ratings, not short duration 'boost' ratings.

Support regenerate break, SOC algorithm , magnetic eddy current brake control, Emergency stop-and-reverse motor control, Anti-sliding slope.

Indirect Field Orientation (IFO) Vector Control algorithm generates the maximum possible torque and efficiency across the entire speed range.

Enables quick and easy characterization of the AC motor without having to remove it from the vehicle, A5 series controllers are fully compatible with any brand of AC motor.

Auto temperature compensation, remote calibration.

Advanced Space Vector Pulse Width Modulation (SVPWM) techniques produce low motor harmonics, low torque ripple and minimized heating losses, resulting in high efficiency.

Robust Safety and Reliability

Insulated metal substrate power-base provides superior heat transfer for increased reliability.

Patented shoot-through guide architectural improve the shoot-through capability and reliability.

Dual Microprocessor architecture ensures the highest possible functional safety performance level is achieved.

Reverse polarity protection on battery connections.

Short circuit protection on all output drivers.

Comprehensive fault detect function, support over-voltage, under-voltage, over current, overheated and Reverse polarity protection.

Sealed to IP65 .

Simple & Flexible

Custom programmable for different requirements of EV OEM.

PC Windows programming and powerful system diagnostic tools.

Buzzer alrm for instant diagnostic and faulty position.

Supports CANbus and CANopen protocols.

Speci cation

P/N	A5-435X	A5-440X	A5-535X	A5-540X
Nominal Input Voltage (V)	36~48	36~48	48~72	48~72
2 Min output Current Rating (Arms)	350	400	350	400
60 Min output Current Rating (Arms)	150	175	150	175
Sensor	Incremental encoder			
Communication	CAN			
Working Temperature	-30℃~55℃			
Storage Temperature	-40℃~85℃			
Insulation	20MΩ Insulation resistance, 0.05mA leakage @ DC 1000V In/Out on casing.			
IP Rating	IP65			
Efficiency	97%			
Cooling	Self-cooled			
Vibration Standard	GB/T2423			
Control Method	IFO Vector			
Heatsink Requirement	Installed in well-ventilated or an air cooling system is required			
Application	AC Induction Motor			

Note: 60 Min RMS Current Rating is measured under proper heat-sink assistance.

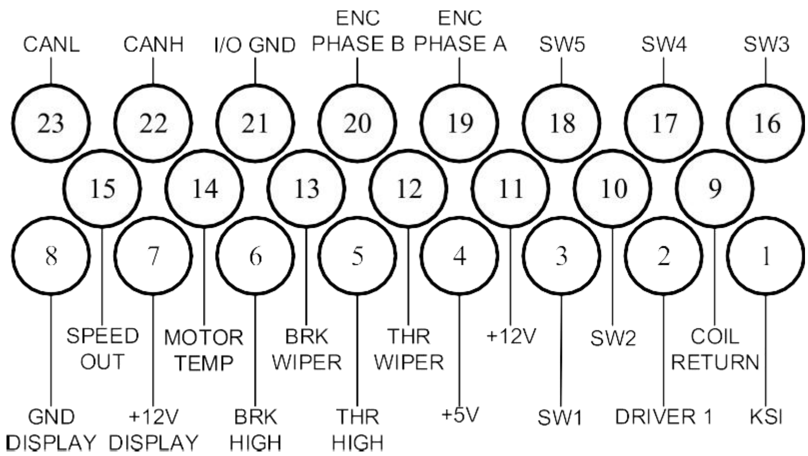
Typical Application

Golf

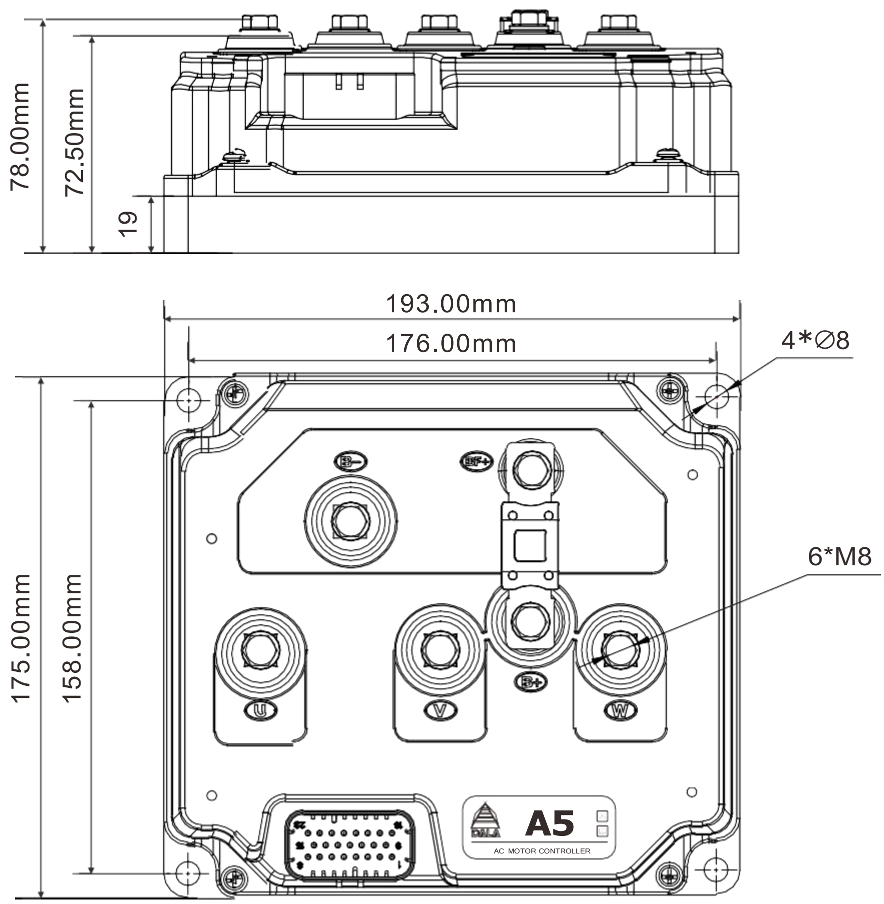
Personnel Carriers

Light on-road

Connector Wiring



DIMENSIONS



Typical Wiring